

4(a)(i)	5.0 cm	A1
4(a)(ii)	$\omega = 2\pi / T$ or $\omega = 2\pi f$ and $f = 1 / T$	C1
	$\omega = 2\pi / 4.0$ $= 1.6 \text{ rad s}^{-1}$	A1
4(a)(iii)	$v_0 = \omega x_0$	C1
	$= 1.57 \times 5.0$ $= 7.9 \text{ cm s}^{-1}$	A1
4(b)	- initial pull was to the right - distance from X to trolley (at equilibrium) is 20 cm - period is 4.0 s - initial motion undamped - motion becomes damped at/from 12 s - damping is light - maximum speed at 1 s, 3 s, etc. / stationary at 2 s, 4 s, etc. <i>Any three points, 1 mark each</i>	B3
4(c)	sketch: closed loop encircling (20, 0)	B1
	minimum L shown as 15 cm <u>and</u> maximum L shown as 25 cm	B1
	minimum v shown as -7.9 cm s^{-1} <u>and</u> maximum v shown as $+7.9 \text{ cm s}^{-1}$	B1